Clinical Dynamics and Therapeutic Countermeasures for Nipah Virus: A Comprehensive Analysis

Authors: Navneet Kaur Bajwa, Parth Mehta, Mukul Saini, Chander Kant
Department of Computer Science Engineering
Chitkara University, Punjab
E-Mails:navneet600.be22@chitkara.edu.in,parth644.be22@chitkara.edu.in,
mukul586.be22@chitkara.edu.in,chander1455.be22@chitkara.edu.in

Abstract— The Nipah virus (NiV) is a highly infectious zoonotic pathogen that can cause potentially fatal central nervous system and respiratory diseases. The current absence of targeted antiviral medicines and preventative vaccines for both the treatment of NiV and the management of outbreaks presents a major challenge to managing any potential future outbreaks of this pathogen. Recent changes in the epidemiology of NiV, including the 2025−2026 spillover events occurring in Bangladesh and India, greatly increase the urgency to evaluate and test potential medical countermeasures against NiV (e.g. Hu1F5 MAb and ChAdOx1 NipahB vaccine) in clinical settings. This paper provides a systematic review of the molecular pathogenesis of NiV, the different phenotypes of the various genetic strains, the temporal viral kinetics, and the latest epidemiological data regarding the virus. The incorporation of these findings will support the development of future clinical trials; they also underscore the importance of developing uniform care standards, as well as integrated regional surveillance networks, in order to support timely and effective outbreak responses.

Keywords—Nipah virus, paramyxoviridae, viral pathogenesis, monoclonal antibodies, outbreak surveillance, clinical trials.

I. INTRODUCTION

Nipah virus (NIV) is a highly infectious zoonotic pathogen classified as a member of the paramyxovirus family (genus Henipavirus) and is a globally acknowledged priority pathogen with significant potential to cause negative public health consequences.[1] Initial identification of the virus occurred in 1998 in Malaysia and Singapore, where the primary intermediate hosts (swine) were identified. Since then, the epidemiological footprint of the virus has moved to India and surrounding countries.[2] In recent outbreaks in Bangladesh and India, the primary mode of criminal transmission has been direct bat-to-human contact, followed by efficient human-to-human transmission.[4]

NiV infection has a very high CFR typical values between 40 - 75%, sometimes higher in individual outbreak or cluster locations.[1] Management at this time consists only of empirical supportive therapy[1]. The research for next-generation treatments and vaccines is advancing; however, any clinical trials will need a thorough understanding of the disease's natural history.[2] The most recent outbreaks (2025 - 2026) in the locations included Kerala, West Bengal, and Bangladesh suggest that there is an urgent need to turn experimental countermeasures into an active intervention.[9] The purpose of the analysis contained within this paper is to evaluate and optimize existing methodology as future therapeutic trials undergo testing.[12]

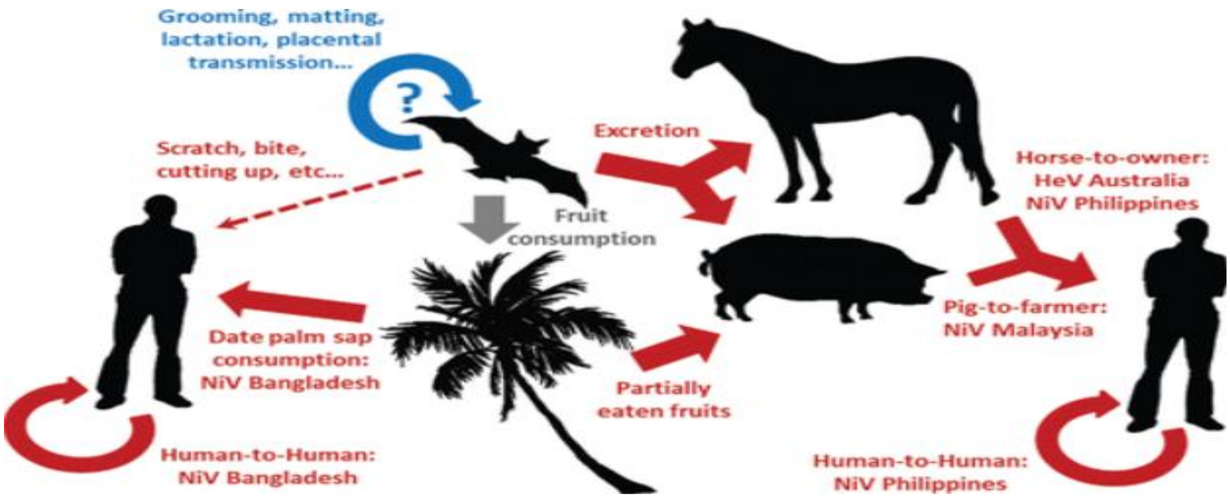

Figure 1. Transmission cycle of Nipah virus showing zoonotic spillover from *Pteropus* bats to humans and subsequent human-to-human transmission via respiratory droplets and close contact.'

II. METHODOLOGY

For this analysis, a systematic review has been conducted of epidemiological, clinical, and genomic data regarding Nipah virus infection. These data have been obtained from peer-reviewed publications (PubMed, BMJ Open, Nature), WHO outbreak reports, and national surveillance systems (NCDC, ICMR).

The following are the inclusion criteria used for studies to be included in the review:

- Publication dates ranging from 2015-2026
- At least 100 confirmed cases in the clinical dataset
- Genomics studies with validated sequencing data

The analysis of data will use:

- Comparative strain analysis between NiV-M and NiV-B strains.
- Aggregated clinical symptom evaluation of 717 cases across the total number of cases analysed.
- Epidemiological Trend/Analysis.

III. VIROLOGY AND CELLULAR PATHOGENESIS

NiV is an enveloped virus containing a single-stranded, negative-sense RNA genome of approximately 18.2 kilobases.[1] This genome encodes six major structural proteins responsible for replication and cellular entry.[1] Pathogenesis is heavily dictated by the attachment (G) and fusion (F) glycoproteins.[1] The G protein binds with high affinity to ephrin-B2 and ephrin-B3 receptors, which are ubiquitously expressed on endothelial cells and respiratory and neuronal tissues, ensuring the infection is systemic rather than localized.[1]

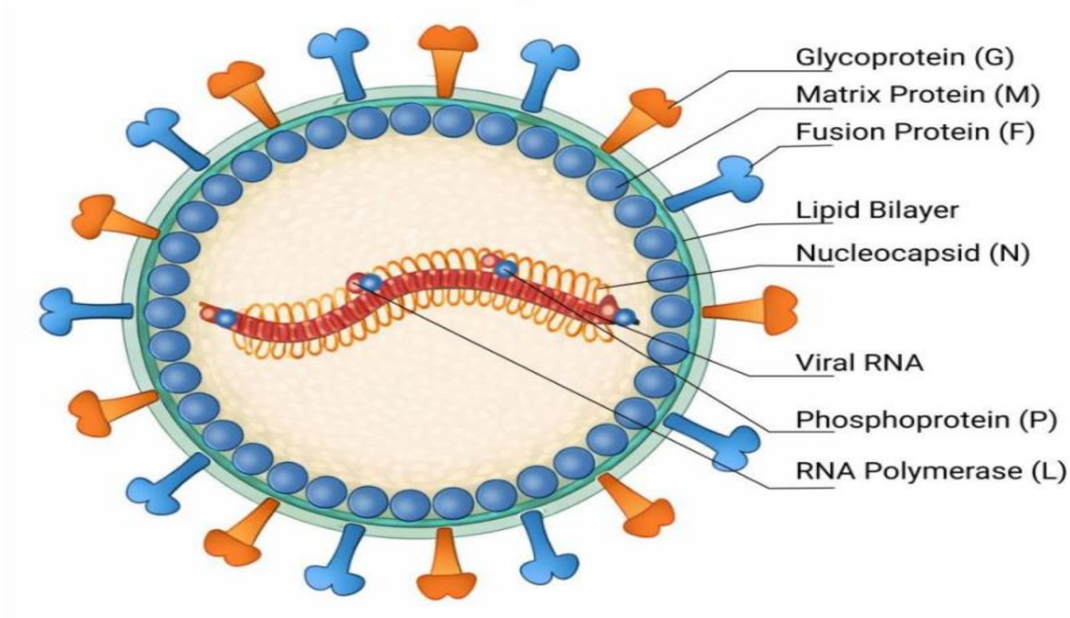

Figure 2. Structural organization of Nipah virus highlighting the RNA genome (~18.2 kb), envelope, and surface glycoproteins (G and F) responsible for host cell attachment and fusion.

Following receptor binding, the F glycoprotein undergoes a conformational change to facilitate membrane fusion and viral entry.[15] Once inside the host cell, products of the viral P gene actively suppress host interferon signaling, enabling rapid viral replication.[1] The defining pathological feature of NiV is extensive microvascular destruction, which triggers thrombosis, ischemia, and severe inflammation within the lungs and the central nervous system (CNS).[1] Additionally, the virus can persist in immune-privileged CNS compartments, leading to relapsing or late-onset encephalitis.[1]

IV. CLINICAL PHENOTYPES AND STRAIN DIVERGENCE

A. Dual CNS and Respiratory Tropism

An aggregated analysis of 717 laboratory-confirmed cases indicates that there are two pathologies involved in the disease process caused by NiV in both the Central Nervous System (CNS) and respiratory tract.[2] Fever at the onset of symptoms/severity occurs in 99% of persons with NiV infection.[2] Rapidly progressing neurologic deterioration occurs and can be characterized by: headache (70% of cases); confusion (65%); altered level of consciousness (62%).[2] There is frequently an acute microinfarct on Magnetic Resonance Imaging (71%).[2] Respiratory complications were also present in a subset of the cases, with cough noted in 45%; difficulty breathing in approximately 58%.

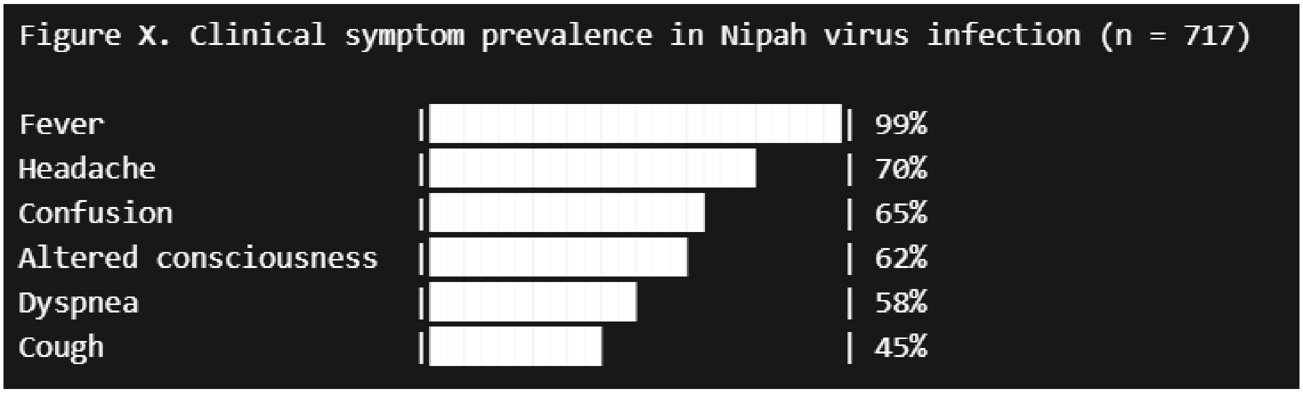

Figure 3. Clinical symptom prevalence in Nipah virus infection (n = 717 cases).

B. Phenotypic Variations

There are notable differences in the phenotypes of the Malaysian (NiV-M) and Bangladeshi (NiV-B) clades. The NiV-B infection has a higher rate of respiratory events than the NiV-M infection (e.g., cough 49% vs 16%, shortness of breath 51% vs 2%).[2] In contrast, neurological symptoms related to the infection (e.g., myoclonus) are more common in NiV-M cases.[2] The respiratory tropism of the NiV-B strain makes person-to-person transfer through aerosols more efficient.[2]

Table I	**Phenotypic Divergence Between NiV Clades**
Clinical Manifestation	Prevalence in NiV-M Strain
Cough	16% (17/105)
Shortness of Breath	2% (2/103)
Fatigue / Malaise	20% (21/104)
Vomiting	36% (41/114)
Anorexia	27% (28/103)

Altered Consciousness	23% (23/101)
Headache	87% (99/114)
Myoclonus	51% (56/110)

V. VIRAL KINETICS AND THE THERAPEUTIC WINDOW

For NiV, the typical incubation period lasts between 3-14 days with rare outliers lasting up to 45 days.[2] The presence of viral RNA can be found via oral swabs starting the day after the onset of symptoms.[2] Systemic viremia, however, has a very narrow window of peak levels between 4-10 days after the start of symptoms.[2] CNS involvement generally occurs from 5-7 days after onset of symptoms.[2] Because of this timing, there is only a narrow window for administering antiviral therapeutics (early in the viremic phase) and preventing significant irreversible tissue necrosis.[2]

Age, degree of neurological depression, severity of autonomic dysregulation, and baseline viral load all have a strong association with mortality from the virus.[2] There is large variation in the quality of baseline supportive care across regions (making it difficult to evaluate therapy), so any future clinical trials should use a strictly enforced standardized treatment protocol to ensure the integrity of the data collected.[2]

VI. EPIDEMIOLOGICAL EVOLUTION (2025–2026)

A. Geographic Expansion and Nosocomial Clusters

Recurrent clusters of viral transmission in Kerala during May to July 2025 demonstrated the geographic expansion of the virus into previously unaffected districts (i.e., Palakkad).[6] This viral expansion has been traced to independent zoonotic spillover events rather than to undetected human transmission chains.[22] In January 2026, there were two nosocomial infections among nurses (i.e., in a health care setting) associated with a large outbreak in West Bengal. Epidemiological tracing identified that both nurses[6] had contact with a currently undiagnosed index patient presenting with acute respiratory distress, illustrating how atypical clinical presentations can evade detection by standard encephalitis surveillance systems and result in amplification of disease at the facility level.[25]

Location	Year	Cases	Deaths	CFR
Malaysia	1998–99	276	106	38%

Bangladesh	2001–2026	300+	200+	~67%
Kerala	2018	23	21	91%
West Bengal	2026	Cluster	Fatal	High

B. Genomic Dynamics

In Bangladesh, there were numerous incidences of sporadic zoonotic spillovers in early 2026 which reconfirmed that drinking raw date palm sap is a risk factor.[10] Genomic sequencing of the most recent isolates originating from eastern India confirmed their cluster in the NiV-B clade, exhibiting high nucleotide identity (~99.2%) to historical Bangladesh isolates and exhibiting a ~9% divergence from NiV-M clade.[5] There is speculation that the mutations occurring primarily in the P, F and L genes may be responsible for the high levels of respiratory tropism and virulence seen with the NiV-B clade.[14]

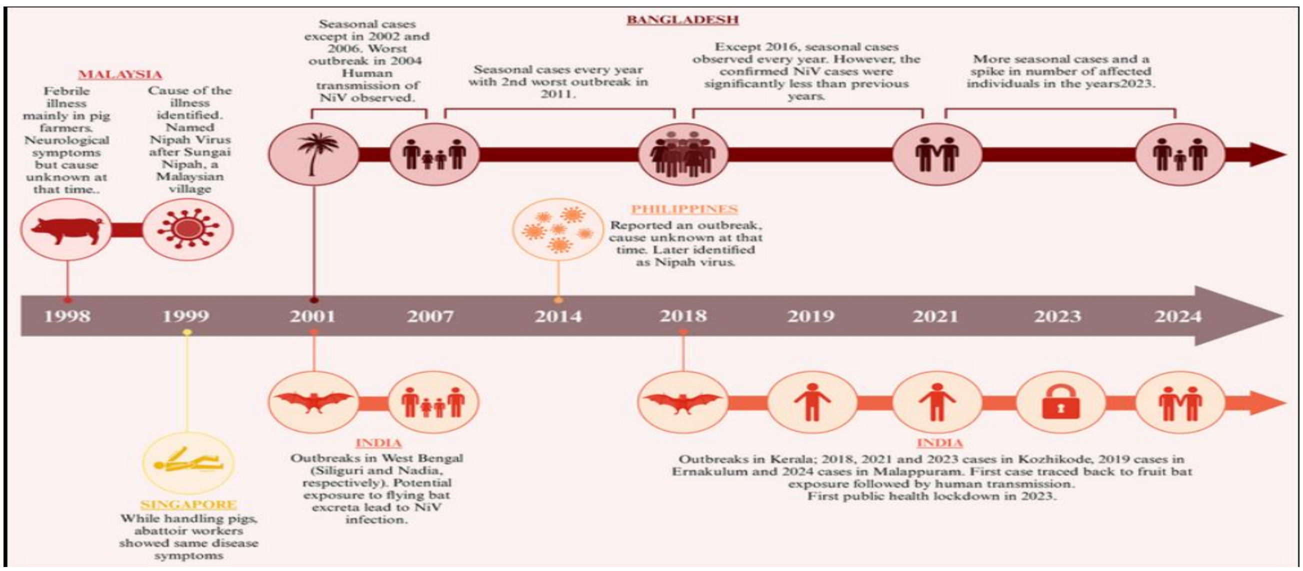

Figure 4. Timeline of Nipah virus outbreaks across Malaysia, Singapore, India, Bangladesh, and the Philippines (1998–2024), showing transmission patterns, seasonal trends, and major public health responses.

VII. MEDICAL COUNTERMEASURE PIPELINE

A. Monoclonal Antibodies

Post-exposure Prophylaxis has fundamentally changed with the invention and development of a new monoclonal antibody, Hu1F5. While other therapies are directed toward the target F glycoprotein after the F glycoprotein has undergone a conformational change to its pre-fusion state, Hu1F5 recognizes and binds with the F glycoprotein in its pre-fusion state to bind and stop the viral entry and prevent the destructive cellular syncytia making.[15] Non-human primate studies have shown a 100% survival rate with Hu1F5 administered as late as Days Post Infection.[15] Accelerated Phase 1b/2a clinical trial study supported by CEPI will begin in India and Bangladesh.[13]

Feature	Hu1F5 Antibody	ChAdOx1 NipahB Vaccine
Type	Passive	Active

Timing	Post-exposure	Pre-exposure
Evidence	100% survival (primates)	Phase II trials
Limitation	Short-term	Delayed immunity

B. Vaccine Architectures

The ChAdOx1 NipahB vaccine was first tested in Phase II clinical trials in December 2025 by the University of Oxford in Bangladesh.[12] This vaccine candidate uses a replication-deficient chimpanzee adenovirus vector to present NiV antigens safely.[12] In addition, the Serum Institute of India (SII) is collaborating with the University of Oxford to create an investigational ready-reserve of 100,000 doses of the vaccine. This allows for the immediate implementation of ring vaccination strategies if an outbreak of NiV occurs.[40]

VIII. CLINICAL TRIAL ARCHITECTURE AND SURVEILLANCE

A. Defining Endpoints and Diagnostics

The Bangladesh Acute Encephalitis Syndrome (BASE) cohort is currently recruiting longitudinal clinical data from a total of 2,000 patients.[12] The follow-up protocols being employed by the BASE study extend to a duration of 90 days, in order to investigate why so many (up to 26%) of Nipah virus (NiV) survivors will have long-term chronic neurological deficits. With the increasing number of patients being recruited into this study, future clinical trials will be able to assess treatment effects based on long-term outcomes rather than solely on survival during the acute phase after infection with NiV.[2]

In addition, the BASE study is currently prioritizing implementation of decentralized rapid diagnostic testing methods (e.g., RT-LAMP assays) in order to eliminate testing bottlenecks that impede timely patient enrollment (e.g., time required to obtain laboratory results).[51] Interventions that are temperature-stable are needed in order to provide a target product profile (TPP), as well as maximize central nervous system (CNS) penetration at a rapid rate.[53]

B. The One Health Imperative

Since outbreaks can have major regional effects both economically and diplomatically, isolated human actions are not enough to solve the problem.[4] A comprehensive "One Health" approach is necessary to prevent the first spillover event. This type of approach is characterized by a combination of good virological surveillance (i.e., monitoring of bat reservoirs) and effective agricultural practice modifications designed to reduce the frequency of contacts between people and wildlife.[4]

IX. CONCLUSION

Nipah virus (NiV) is an extremely pathogenic organism due to its ability to infect multiple cell types (i.e., dual tropism), its ability to cause rapid clinical decline in patients, and its ability to undergo rapid antigenic variation. Researchers face challenges such as limited therapeutic windows and accessibility to supportive care when transitioning experimental therapies such as Hu1F5 antibody and ChAdOx1 Nipah B vaccine to clinical samples for evaluation. To produce reliable efficacy data from these therapeutic agents and improve global readiness for future pandemics, standardization of clinical protocols, increasing the number of longitudinal endpoint assessments, and continued investment in decentralized diagnostic testing are critical components of successful development of therapeutic agents to combat NiV.

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